Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/05/2010
PI Name:	Wang, Ya M.D., Ph.D.		
Project Title:	The mechanism of excess relative risk on carcinogenesis induced by high-LET radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30322	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Guida, Peter Ph.D. (BNL)		
Grant/Contract No.:	NNX09AF24G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our hypothesis in this application is that high LET radiation induced excess relative risk on mutagenesis as well as on carcinogenesis is due to inefficient Ku-dependent non-homologous end-joining (NHEJ) repair. The results from this project will provide a mechanism to explain why high LET radiation induced such excess relative risk (ERR), and will significantly improve our understanding of the relationship between DNA repair and high LET radiation induced such ERR. Therefore, the results from this proposal will assist NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating high LET radiation induced ERR on mutagenesis as well as on carcinogenesis and by providing protective targets for reducing high LET radiation induced mutagenesis as well as carcinogenesis.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study described in this application will determine whether the major reason for high linear energy transfer (LET) ionizing radiation (IR) to induce excess relative risk (ERR) as compared with low LET IR on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining repair. The results from this proposal will answer the question whether inefficient Ku-dependent repair is the major reason for high LET IR to induce ERR on carcinogenesis. Results from this proposal will assist the NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating the ERR on carcinogenesis and by providing specific protective targets for reducing high LET IR-induced ERR on the carcinogenesis.
Task Progress:	Progress and Impact: This project includes two major categories: Animal experiment and cells experiment. We report the progress from these two categories. 1. Animal work: Although the project progress was affected due to the PI's laboratory relocation, (for example, we could not make the BNL 2008 fall run because of waiting for animal protocols to be approved) we still managed to initiate irradiating the cells and mice in BNL in spring and fall of 2009. The animal work requires several BNL beam times, which is planned to be finished in 2010. 2. Cells work: We carried out some experiments by using different types of cells and obtained some interesting results: 1). S-phase cells are more sensitive to heavy ion, the high linear energy transfer (LET) radiation (component of space radiation), although they are more resistant to x or gamma ray. After further study of the mechanism, we believe that the phenotype of S-phase sensitive to high-LET radiation is due to the fact: high-LET radiation could generate much more small fragments (< 40 base pairs) that result in inefficient Ku binding in S –phase, which further supports our hypothesis: high-LET killing more cells when compared with low-LET radiation is because of inhibition of Ku-dependent NHEJ. 2). We previously reported that DNA double strand break (DSB) activated checkpoint mainly facilitates homologous recombination repair (HRR) and has little effect on non-homologous end-joining (NHEJ). Combining our new finding with high-LET radiation we tested our hypothesis that checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells and obtained supportive results. 3). It has been know for more than two decades that low dose radiation (< 0.1 Gy) could induce adaptive responses. The adaptive responses could protect cells and animals from high challenge dose induced killing and carcinogenesis. Because DNA DSBs induced by radiation are the severest threat for cell survival and genetic integrity, the low dose adaptive responses are believed to promote DNA repair. However, it remains unclear which DNA DSB repair pathway: NHEJ or HRR is promoted by the low dose radiation. We found that low dose radiation could not induce any protective role in high-LET irradiated cells. After combing these results with our repair theory about high-LET radiated cells, we hypothesized that low dose radiation induced adaptive responses are through promoting NHEJ. We tested this hypothesis and obtained supportive results. These results further support our hypothesis that the major reason for high LET IR to induce excess relative risk on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining (NHEJ). Because our results are provide mechanism information to elucidate the nature different between high-LET radiation and low-LET radiation, these results will contribute to both estimating and reducing the risk of space radiation to the astronauts' health. Bibliography: Abstracts for Proceedings: 1. Wang, H. Yu, X. and Wang, Y. A small peptide protects cells from high-LET radiation-induced damage. Heavy ions in therapy and space symposium. Cologne, Germany July 6-10, 2009. 2. Yu, X., Wang, H., Wang, P. and Wang, Y. Ku dependent NHEJ pathway contributes to low dose radiation induced adaptive responses. 55th Annual Meeting of the Radiation Research Society, Savannah, GA, October 3-7, 2009. 3. Wang, H., Yu, X. Zhang, X. and Wang Y. A small peptide protects cells ionizing radiation induced killing. 55th Annual Meeting of the Radiation Research Society, Savannah, GA, October 3-7, 2009. Articles in Peer-Reviewed Journals: 1. Wang, H., Wang, X. Zhang, PY and Wang, Y. The Ku-dependent non-homologous end-joining but not other repair pathway is inhibited by high-linear energy transfer ionizing radiation. DNA Repair, 7: 725-733, 2008. 2. Wang, H. and Wang, Y. Checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells. Cell Cycle. 7: 2444-2445, 2008. 3. Wang, H., Liu, S., Zhang, P-Y, Zhang, S-M, Naidu, M., Wang, H-C and Wang, Y. S-phase cells are more sensitive to high linear energy transfer radiation. IJROBP, 74:1236-1241, 2009. 4. Wang, H. and Wang, Y. Enhancing sensitization of S-phase cells to high-linear energy transfer radiation by inhibition of homologous recombination repair. Cell Cycle, 8: 1451-1452, 2009 5. Yu, X., Wang, H., Wang, P., Guida, P., Chen, B.P.C. and Wang, Y. The Ku dependent non-homologous end-joining pathway contributes to low dose radiation-stimulated cell survival Submitted, 2009 6. Wang, H., Zhang, X., Wang, P., Yu, X., Essers, J., Chen, D., Kanaar, R., Takeda, S. and Wang, Y. Characteristics of DNA binding proteins determine the biological sensitivity to high linear energy transfer radiation. Submitted, 2009.
Bibliography Type:	Description: (Last Updated: 07/07/2021)
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Articles in Peer-reviewed Journals	Wang H, Wang Y. "Enhancing sensitization of S-phase cells to high-linear energy transfer radiation by inhibition of homologous recombination repair." Cell Cycle. 2009 May 1;8(9):1451-2. PubMed PMID: 19342898 , May-2009